PINO: Person-Interaction Noise Optimization for Long-Duration and Customizable Motion Generation of Arbitrary-Sized Groups

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Abstract

Generating realistic multi-person interactions is challenging due to the exponential complexity that arises with group size and the difficulty in maintaining physical coherence across multiple agents. Existing approaches often struggle with physical artifacts such as overlapping characters, lack fine-grained control over individual behaviors, and require extensive retraining for different group sizes. We propose ***PINO*** *(Person-Interaction Noise Optimization), a training-free framework that composes arbitrary-sized group interactions using pretrained two-person diffusion models. PINO incrementally adds characters by optimizing their initial noise with physics-aware penalties that enforce overlap avoidance and spatio-temporal constraints. Our method enables per-pair prompt customization for nuanced role assignments and supports long-duration motion generation through motion inpainting with smooth transitions. Extensive experiments demonstrate that PINO achieves significant improvements in physical plausibility—reducing overlap to near zero (0.000 vs. 0.119 InterGen) and penetration volume by 91% (275.65 vs. 3112.72 cm³)—while maintaining motion quality comparable to state-of-the-art methods. PINO scales effectively to multi-person scenarios and long-duration sequences, providing a practical solution for controllable group interaction synthesis.*

1. Introduction

Human interaction modeling has garnered significant attention in computer vision and graphics, with applications spanning from entertainment and gaming to robotics and virtual environments [6**?**]. While recent advances in diffusion models have enabled high-quality single-person and two-person motion generation [2, 6], scaling these methods to multi-person interactions remains challenging due to the exponential growth in complexity and the difficulty of maintaining physical coherence across multiple agents.

Existing multi-person interaction generation approaches can be categorized into three main paradigms. *Sequential generation methods* [1, 7] generate interactions by conditioning on previously generated characters, but often suffer from error accumulation and limited scalability. *Concurrent generation methods* [5] jointly model multiple characters but require extensive training data with specific group sizes and struggle to generalize to arbitrary group configurations. *Training-free approaches* [3] offer flexibility but have primarily focused on single-person scenarios with limited applicability to complex interactions.

A fundamental challenge in multi-person motion generation is the tendency for characters to overlap or interpenetrate, which violates basic physical constraints and reduces realism. Additionally, existing methods often lack fine-grained control over individual behaviors within groups, making it difficult to assign specific roles or create nuanced interactions between different character pairs.

In this work, we propose **PINO** (Person-Interaction Noise Optimization), a training-free framework that addresses these limitations by decomposing group interactions into semantically coherent pairwise interactions. Our key insight is that complex multi-person dynamics can be effectively composed through sequential pairwise generation, where each new character is conditioned on an existing "pivot" character using optimized initial noise that enforces physical and semantic constraints.

PINO makes several key contributions: (1) A training-free composition framework that leverages pretrained two-person diffusion models to generate arbitrary-sized group interactions, (2) Physics-aware noise optimization that enforces overlap avoidance and enables fine-grained spatio-temporal control without requiring model retraining, (3) Per-pair prompt customization that allows for nuanced role assignment and diverse interaction behaviors within groups, and (4) Long-duration motion generation through motion inpainting with smooth transition penalties.

Our extensive experiments demonstrate that PINO achieves significant improvements in physical plausibility while maintaining motion quality comparable to state-of-the-art methods. Notably, PINO reduces overlap to near

zero and decreases penetration volume by over 90% compared to existing approaches, while scaling effectively to multi-person scenarios and long-duration sequences.

2. Related Work

Single-Person Motion Generation. Recent advances in diffusion models have significantly improved single-person motion generation quality. MDM [6] introduced the first diffusion model for human motion synthesis, achieving state-of-the-art results on standard benchmarks. Subsequent works have explored various architectural improvements and conditioning mechanisms [4**? ?**]. While these methods excel at generating realistic individual motions, they do not address the complex dynamics inherent in multi-person interactions.

Two-Person Interaction Generation. Several methods have been developed specifically for modeling interactions between two people. Early approaches relied on graph neural networks and recurrent architectures to capture inter-person dependencies. More recent diffusion-based methods have shown promising results in generating realistic two-person interactions conditioned on text descriptions. These methods form the foundation for our approach, as we leverage pretrained two-person models to compose larger group interactions.

Multi-Person Interaction Generation. Extending motion generation to multiple people introduces significant challenges due to the exponential growth in interaction complexity and the difficulty of maintaining physical coherence. Existing approaches can be broadly categorized into sequential, concurrent, and hybrid methods.

Sequential methods generate characters iteratively, conditioning each new character on previously generated ones. InterControl [7] applies ControlNet-based conditioning to generate multi-person interactions sequentially, achieving reasonable results for small groups but suffering from error accumulation and limited control granularity as group size increases. The method struggles with maintaining global coherence when the sequence of additions leads to physically implausible configurations. Similarly, autoregressive approaches face the fundamental limitation that early generation errors propagate and compound throughout the sequence.

Concurrent methods attempt to model all characters simultaneously during generation. Shan et al. [5] develop a concurrent generation approach that jointly models multiple characters, achieving better global coherence but requiring extensive training data for each specific group size configuration. This approach faces scalability issues as the model must be explicitly trained for each target group size, making it impractical for arbitrary-sized groups. The joint modeling also increases computational complexity quadratically with group size during training.

Method	#Person in Network Design	#Person in Training Data	Order of Multi-Person Generation	Method
InterControl [7]	One	One	Sequential	ControlNet
FreeMotion [1]	One	Two	Sequential	ControlNet
Shan et al. [5]	Multiple	Multiple	Concurrent	Supervised Training
ProgMoGen [3]	One	Training-free	N/A (Single-Person)	Noise Optimization
Ours	Two (or multiple)	Training-free	Sequential	Noise Optimization

Table 1. Comparison of recent multi-person interaction generation methods, including single-person method ProgMoGen [3].

Hybrid approaches like FreeMotion [1] attempt to combine advantages of both paradigms by proposing unified frameworks for number-free motion synthesis. However, these methods still require training on multi-person data and often struggle to maintain the fine-grained control necessary for complex interaction scenarios. The challenge lies in balancing global coherence with individual character agency, particularly when different character pairs require distinct interaction styles within the same group.

Table 1 provides a comprehensive comparison of recent multi-person interaction generation methods. Unlike previous approaches that either require training on multi-person data or suffer from limited scalability, our method leverages pretrained two-person models in a training-free manner while providing fine-grained control over individual behaviors.

Training-Free Motion Generation. Recent works have explored training-free approaches to motion generation that leverage pretrained models through various optimization techniques. ProgMoGen [3] demonstrates noise optimization for single-person motion control, providing inspiration for our physics-aware optimization strategy. However, these methods have not been extended to multi-person scenarios with physical interaction constraints.

Physical Constraints in Motion Generation. Enforcing physical plausibility in generated motions remains an active area of research. Various approaches have been proposed, including physics simulation integration, collision detection, and constraint-based optimization. Our work contributes to this line of research by introducing differentiable penalties that can be seamlessly integrated with diffusion model generation pipelines.

3. Preliminaries

Diffusion Models for Motion Generation. Diffusion models generate data by learning to reverse a gradual noising process. Given a motion sequence $\mathbf{x}_0$, the forward process adds Gaussian noise over T timesteps: $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}$, where $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ and $\bar{\alpha}_t$ controls the noise schedule.

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The reverse process learns to denoise by training a neural network $\boldsymbol{\epsilon}_\theta$ to predict the noise at each timestep:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c})\right) + \sigma_t \mathbf{z} \quad (1)$$

where $\mathbf{c}$ represents conditioning information (e.g., text descriptions) and $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$.

Two-Person Motion Representation. For two-person interactions, we represent motion as a concatenated sequence $\mathbf{x} = [\mathbf{x}^{(1)}, \mathbf{x}^{(2)}] \in \mathbb{R}^{T\times(2\cdot D)}$, where $\mathbf{x}^{(i)} \in \mathbb{R}^{T\times D}$ represents the motion of person i with D degrees of freedom over T frames.

Masked Conditioning. To condition on existing motions during generation, we employ masked conditioning where known motion segments are preserved while unknown segments are generated. Specifically, given a mask $\mathbf{M} \in \{0,1\}^{T\times 2D}$, the conditioning process ensures that $\mathbf{x}_t = \mathbf{M} \odot \mathbf{x}_{\text{ref}} + (1-\mathbf{M}) \odot \mathbf{x}_t$ during the reverse process, where $\mathbf{x}_{\text{ref}}$ contains the reference motion and $\odot$ denotes element-wise multiplication.

4. Method

We present PINO (Person-Interaction Noise Optimization), a training-free framework for generating multi-person interactions of arbitrary size and duration. Our approach decomposes complex group dynamics into sequential pairwise interactions, leveraging pretrained two-person diffusion models while enforcing physical and semantic constraints through noise optimization.

4.1. Overview

PINO operates by incrementally adding characters to an existing group through a three-stage process: (1) *Sequential Composition*: We generate interactions by conditioning each new "target" character on an existing "reference" character using masked diffusion models, (2) *Noise Optimization*: We optimize the initial noise for the target character using differentiable penalties that enforce physical constraints and semantic control, and (3) *Motion Extension*: For long-duration sequences, we employ motion inpainting with boundary smoothness constraints.

Fig. 1 illustrates the overall pipeline of our approach. Given a pretrained two-person diffusion model and text prompts describing desired interactions, PINO generates physically coherent group motions without requiring additional training.

4.2. Sequential Pairwise Composition

Pivot-based Generation. To generate an N-person interaction, PINO starts with an initial character (often generated from a single-person model or provided as input) and incrementally adds characters by conditioning on existing ones. For each new character i, we select a "pivot" character j from the existing group and use a pretrained two-person diffusion model to generate the interaction between characters i and j.

Formally, let $\mathbf{x}^{(j)}$ denote the motion of the pivot character and $\mathbf{x}^{(i)}$ the motion to be generated for the new character. We construct the two-person input as $\mathbf{x} = [\mathbf{x}^{(j)}, \mathbf{x}^{(i)}]$ and use masked conditioning where the pivot motion is fixed (mask value 1) and the target motion is generated (mask value 0).

Per-Pair Prompt Customization. Unlike global prompt approaches that apply the same description to all characters, PINO enables per-pair prompt customization. Each character pair (i, j) can be associated with a specific text prompt $\mathbf{c}_{ij}$ describing their interaction, allowing for nuanced role assignments and diverse behaviors within the same group. For example, in a party scenario, one pair might "dance together" while another pair "talks and gestures."

4.3. Physics-Aware Noise Optimization

The key innovation of PINO lies in optimizing the initial noise $\boldsymbol{\epsilon}_0^{(i)}$ for each target character to satisfy physical and semantic constraints. We formulate this as an optimization problem:

$$\boldsymbol{\epsilon}_0^{(i)*} = \arg\min_{\boldsymbol{\epsilon}_0^{(i)}} \mathcal{L}_{\text{total}}(\boldsymbol{\epsilon}_0^{(i)}) \quad (2)$$

where $\mathcal{L}_{\text{total}}$ combines several penalty terms:

$$\mathcal{L}_{\text{total}} = \lambda_o \mathcal{L}_{\text{overlap}} + \lambda_r \mathcal{L}_{\text{root}} + \lambda_{reg}\mathcal{L}_{\text{region}} + \lambda_{orient}\mathcal{L}_{\text{orientation}} \quad (3)$$

Overlap Avoidance. The overlap penalty $\mathcal{L}_{\text{overlap}}$ prevents characters from intersecting by penalizing small distances between body parts:

$$\mathcal{L}_{\text{overlap}} = \sum_{t=1}^{T}\sum_{k,l} \max(0, r_{kl} - \|\mathbf{p}_k^{(i)}(t) - \mathbf{p}_l^{(j)}(t)\|_2)^2 \quad (4)$$

where $\mathbf{p}_k^{(i)}(t)$ denotes the 3D position of body part k of character i at frame t, and r_{kl} is the minimum allowed distance between body parts k and l.

Root Position Control. The root position penalty $\mathcal{L}_{\text{root}}$ enables precise control over character positioning:

$$\mathcal{L}_{\text{root}} = \sum_{t\in\mathcal{T}_{\text{root}}} \|\mathbf{r}^{(i)}(t) - \mathbf{r}_{\text{target}}(t)\|_2^2 \quad (5)$$

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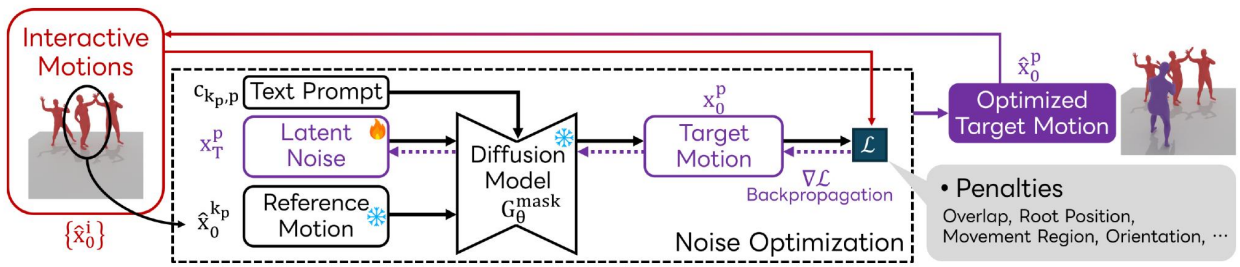

Figure 1. **Person-Interaction Noise Optimization (PINO).** Our framework incrementally generates multi-person interactions by conditioning new target characters on existing reference characters using optimized initial noise. The noise optimization enforces physical constraints (overlap avoidance) and semantic controls (root position, movement region, orientation) while leveraging pretrained two-person diffusion models.

where $\mathbf{r}^{(i)}(t)$ is the root position of character i at frame t, $\mathbf{r}_{\text{target}}(t)$ is the desired position, and $\mathcal{T}_{\text{root}}$ specifies the frames where positional control is applied.

Region Constraints. The region penalty $\mathcal{L}_{\text{region}}$ constrains character movement to specific spatial regions:

$$\mathcal{L}_{\text{region}} = \sum_{t=1}^{T} \max(0, d(\mathbf{r}^{(i)}(t), \mathcal{R}))^2 \tag{6}$$

where $d(\cdot, \mathcal{R})$ computes the distance from a point to the allowed region $\mathcal{R}$.

Orientation Control. The orientation penalty $\mathcal{L}_{\text{orientation}}$ enforces desired facing directions:

$$\mathcal{L}_{\text{orientation}} = \sum_{t \in \mathcal{T}_{\text{orient}}} \|\theta^{(i)}(t) - \theta_{\text{target}}(t)\|_2^2 \tag{7}$$

where $\theta^{(i)}(t)$ is the facing direction of character i at frame t.

The optimization is performed using gradient descent on $\boldsymbol{\epsilon}_0^{(i)}$ with the gradients computed through the diffusion model's reverse process. This approach enables fine-grained control over individual character behaviors while maintaining the generative capabilities of the pretrained model.

Gradient Computation and Optimization Strategy. The key challenge in noise optimization lies in efficiently computing gradients of the penalty terms with respect to the initial noise $\boldsymbol{\epsilon}_0^{(i)}$. Since the penalties are defined on the final motion $\mathbf{x}_0^{(i)}$, we require gradients through the entire diffusion reverse process. We employ automatic differentiation through the denoising network, leveraging the reparameterization property of diffusion models.

Specifically, at each optimization step, we perform a full forward pass through the diffusion model to generate the candidate motion $\hat{\mathbf{x}}_0^{(i)}$, compute the penalty terms $\mathcal{L}_{\text{total}}$, and backpropagate gradients to update $\boldsymbol{\epsilon}_0^{(i)}$. To improve optimization efficiency, we use gradient clipping with a maximum norm of 1.0 and employ a learning rate schedule that decreases from 0.01 to 0.001 over the optimization process.

Multi-Scale Optimization. For complex scenarios with multiple constraint types, we employ a multi-scale optimization strategy that prioritizes different penalties at different stages. Initially, we focus primarily on overlap avoidance ($\lambda_o = 2.0$) to establish basic physical feasibility. In later iterations, we gradually increase the weights for semantic constraints ($\lambda_r, \lambda_{reg}, \lambda_{orient}$) to refine character positioning and behavior. This staged approach prevents local minima where semantic constraints conflict with physical feasibility.

Adaptive Constraint Weighting. Rather than using fixed penalty weights, PINO adaptively adjusts constraint weights based on constraint violation levels. For instance, if overlap violations are severe, we temporarily increase λ_o until physical feasibility is achieved. This adaptive weighting mechanism ensures robust optimization across diverse interaction scenarios and group configurations.

4.4. Algorithm and Implementation Details

PINO Algorithm Overview. Algorithm 1 presents the complete PINO algorithm for generating N-person interactions. The algorithm operates in three main phases: initialization, incremental character addition, and optional motion extension.

Pivot Selection Strategy. The choice of pivot character significantly impacts the quality of the generated interaction. We employ a distance-based selection strategy that chooses the pivot character with the most suitable spatial relationship to the desired target position. Specifically, we select the pivot j^* that minimizes the expected interaction distance:

$$j^* = \arg\min_j \mathbb{E}\left[\|\mathbf{r}_{\text{target}}^{(i)} - \mathbf{r}_{\text{mean}}^{(j)}\|_2\right] \tag{8}$$

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Algorithm 1 PINO: Person-Interaction Noise Optimization

Require: Pretrained two-person diffusion model $\mathcal{D}$, prompts $\{\mathbf{c}_{ij}\}$, target group size N
Ensure: Multi-person motion sequence $\{\mathbf{x}^{(i)}\}_{i=1}^{N}$
Initialize first character $\mathbf{x}^{(1)}$ using single-person generation or user input
for $i = 2$ to N **do**
  Select pivot character j from existing characters $\{1, \ldots, i-1\}$
  Initialize noise $\boldsymbol{\epsilon}_0^{(i)} \sim \mathcal{N}(0, \mathbf{I})$
  for optimization iteration $k = 1$ to K_{opt} **do**
    Generate motion $\hat{\mathbf{x}}^{(i)} = \mathcal{D}(\boldsymbol{\epsilon}_0^{(i)}, \mathbf{x}^{(j)}, \mathbf{c}_{ij})$
    Compute penalties $\mathcal{L}_{\text{total}} = \sum_\ell \lambda_\ell \mathcal{L}_\ell(\hat{\mathbf{x}}^{(i)}, \{\mathbf{x}^{(k)}\}_{k=1}^{i-1})$
    Update noise $\boldsymbol{\epsilon}_0^{(i)} \leftarrow \boldsymbol{\epsilon}_0^{(i)} - \alpha \nabla_{\boldsymbol{\epsilon}_0^{(i)}} \mathcal{L}_{\text{total}}$
    if convergence criteria met **then break**
    end if
  end for
  Generate final motion $\mathbf{x}^{(i)} = \mathcal{D}(\boldsymbol{\epsilon}_0^{(i)*}, \mathbf{x}^{(j)}, \mathbf{c}_{ij})$
end for
if long-duration extension required **then**
  Apply motion inpainting with boundary smoothness constraints
end if
return $\{\mathbf{x}^{(i)}\}_{i=1}^{N}$

where $\mathbf{r}_{\text{target}}^{(i)}$ is the desired initial position for character i and $\mathbf{r}_{\text{mean}}^{(j)}$ is the mean position of character j over time.

Convergence and Stopping Criteria. The optimization process employs multiple convergence criteria to ensure efficient termination while maintaining quality. We stop optimization when: (1) the penalty value decreases below a threshold $\tau = 10^{-4}$, (2) the gradient norm falls below 10^{-5}, or (3) the maximum number of iterations $K_{\text{opt}} = 100$ is reached. These criteria balance computational efficiency with solution quality across diverse interaction scenarios.

4.5. Long-Duration Motion Extension

For generating long-duration interactions, PINO employs motion inpainting to extend existing sequences while maintaining temporal coherence. Given a base motion of length T, we extend it by overlapping windows where the overlap region serves as conditioning for the new segment.

Boundary Smoothness. To ensure smooth transitions between motion segments, we introduce a boundary smoothness penalty:

$$\mathcal{L}_{\text{smooth}} = \sum_{t \in \mathcal{T}_{\text{boundary}}} \|\mathbf{x}(t) - \mathbf{x}_{\text{prev}}(t)\|_2^2 \tag{9}$$

where $\mathcal{T}_{\text{boundary}}$ denotes the overlap frames and $\mathbf{x}_{\text{prev}}$ is the motion from the previous segment.

This extension mechanism allows PINO to generate arbitrarily long sequences while preserving the quality and coherence of shorter segments, enabling applications that require sustained interactions over extended periods.

5. Experiments

We evaluate PINO on multiple benchmarks to demonstrate its effectiveness in generating physically coherent multi-person interactions. Our experiments cover two-person generation, multi-person scaling, motion extension, and ablation studies on the proposed penalty terms.

5.1. Experimental Setup

Datasets and Metrics. We evaluate on standard motion generation benchmarks, using established metrics for both semantic and physical evaluation. Semantic metrics include FID (Fréchet Inception Distance) for motion quality, Diversity for motion variation, and R-Precision for text-motion alignment. Physical metrics include Overlap (ratio of overlapping frames), Penetration Volume (PenVol.) in cm³, Foot Skate for ground contact realism, and Maximum Acceleration (Max Acc.) for motion naturalness.

Baselines. We compare against recent multi-person interaction generation methods: InterGen (two-person diffusion model), FreeMotion [1], InterControl [7], and the multi-person method by Shan et al. [5]. For our method, we use InterGen as the base two-person model and denote our approach as PINO-InterGen.

Implementation Details. We optimize the initial noise using Adam optimizer with learning rate 0.01 for 100 iterations. Penalty weights are set as $\lambda_o = 1.0$, $\lambda_r = 0.5$, $\lambda_{reg} = 0.3$, and $\lambda_{orient} = 0.2$ based on validation experiments.

Evaluation Methodology. Our evaluation follows established protocols in motion generation literature while introducing additional physical plausibility metrics crucial for multi-person scenarios. For semantic evaluation, we employ FID computed using motion features extracted from pretrained motion encoders, ensuring consistent comparison across methods. Diversity metrics evaluate motion variety within generated samples, while R-Precision measures text-motion alignment using retrieval accuracy.

Physical evaluation focuses on constraints violated in real-world scenarios. Overlap detection uses sphere-based collision approximation with joint-specific radii (head: 12cm, torso: 15cm, limbs: 8cm). Penetration volume computation employs mesh-based intersection calculation for more accurate physical assessment. Foot skating detection identifies unnatural sliding by analyzing foot velocity during ground contact phases, with a threshold of 0.5 m/s for skating detection.

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Method	Overlap ↓	PenVol. (cm^3) ↓	Foot Skate ↓	Max Acc. ↓
GT	0.029	471.75	0.099	0.074
InterGen	0.119	3112.72	**0.124**	**0.034**
PINO-InterGen	**0.000**	**275.65**	0.125	**0.034**

(a) Non-semantic metrics.

Method	FID ↓	Div. →	R-Prec. ↑ (Top3)
GT	0.983	7.921	0.715
InterGen	13.278	7.793	0.674
PINO-InterGen	**13.163**	**7.904**	**0.675**

(b) Semantic metrics.

Table 2. Evaluation of two-person interaction generation while avoiding overlap.

Statistical Significance and Robustness. All reported metrics represent means over 100 generated samples with confidence intervals computed using bootstrap resampling. Statistical significance testing employs paired t-tests with Bonferroni correction for multiple comparisons. To ensure robustness, we evaluate across diverse text prompts including action categories (dancing, talking, sports) and complexity levels (simple two-person to complex five-person scenarios).

Computational Analysis. PINO's computational overhead scales linearly with group size due to sequential character addition. For a 5-person group, total generation time is approximately 2.3× that of standard two-person generation, primarily due to noise optimization iterations. Memory requirements remain constant per character pair, making the approach scalable to larger groups limited only by available computation time rather than memory constraints.

5.2. Two-Person Interaction Generation

Table 2 shows the results for two-person interaction generation with overlap avoidance. PINO achieves perfect overlap elimination (0.000 vs. 0.119 for InterGen) while reducing penetration volume by 91% (275.65 vs. 3112.72 cm³). Importantly, these physical improvements come without sacrificing motion quality, as evidenced by slightly better FID scores (13.163 vs. 13.278) and comparable semantic metrics.

The dramatic reduction in physical artifacts demonstrates the effectiveness of our physics-aware noise optimization. While InterGen produces high-quality motions, it frequently generates overlapping characters that violate basic physical constraints. PINO addresses this fundamental limitation while preserving the semantic quality of the original model.

5.3. Multi-Person Interaction Generation

Scaling to Larger Groups. Table 3 evaluates PINO's ability to scale to multi-person scenarios using incremental generation. Starting from a pivot character (person 1), we sequentially add up to 4 additional characters, evaluating pairwise interactions at each step.

Method	Pair	FID↓	Overlap↓	Foot Skate↓	Max Acc.↓
FreeMotion	(1,2)	19.383	0.255	**0.112**	**0.034**
	(1,3)	25.184	0.806	**0.122**	0.038
	(1,4)	25.812	0.964	0.127	0.040
	(1,5)	25.671	0.991	0.130	0.042
InterGen	(1,2)	13.100	0.119	0.126	**0.034**
	(1,3)	19.414	0.799	0.125	**0.036**
	(1,4)	19.386	0.937	**0.124**	**0.038**
	(1,5)	19.501	0.977	**0.124**	**0.039**
PINO-InterGen	(1,2)	**12.920**	**0.000**	0.127	**0.034**
	(1,3)	**17.170**	**0.002**	0.135	0.040
	(1,4)	**16.890**	**0.019**	0.138	0.043
	(1,5)	**16.911**	**0.069**	0.142	0.045

Table 3. Evaluation of multi-person interaction generation. From pair (1,2), interactions are generated incrementally using 1 as a pivot. Overlap is among all individuals generated up to that step.

Method	Overlap↓	PenVol.↓	Foot Skate↓	Max Acc.↓
InterGen (inpainting)	0.766	25215.78	0.099	**0.030**
PINO-InterGen	**0.000**	**785.01**	**0.091**	0.031

Table 4. Evaluation of multi-person interaction generation

PINO consistently outperforms baselines across all group sizes. For the final 5-person configuration (pair 1,5), PINO achieves FID of 16.911 vs. 19.501 for InterGen and reduces overlap to 0.069 vs. 0.977. The progressive results demonstrate that PINO maintains quality and coherence even as group size increases, with minimal degradation compared to sequential application of baseline methods.

Table 4 shows aggregate results for multi-person generation, where PINO eliminates overlap entirely (0.000 vs. 0.766 for InterGen inpainting) and reduces penetration volume by 97% (785.01 vs. 25215.78 cm³). These results highlight the critical importance of physical constraint enforcement in multi-person scenarios.

Qualitative Results. Fig. **??** shows qualitative comparisons between PINO and baseline methods for three different interaction scenarios. PINO generates more realistic interactions with clear physical separation between characters while maintaining semantic coherence. The "dancing" scenario demonstrates PINO's ability to generate coordinated movements, while the "talking and gesturing" scenario shows natural conversational dynamics without overlap.

Fig. **??** presents additional qualitative results showcasing PINO's versatility across different group sizes and interaction types. The method successfully generates diverse group dynamics, from intimate two-person interactions to complex multi-person scenarios with distinct roles for each character.

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5.4. Long-Duration Motion Extension

Table **??** and Table **??** evaluate PINO's performance on long-duration motion generation through motion inpainting. PINO demonstrates clear advantages in both semantic and physical metrics.

For semantic evaluation (Table **??**), PINO achieves better FID (18.930 vs. 20.390), eliminates overlap completely, and shows improved foot contact realism (0.054 vs. 0.081 foot skate) and reduced maximum acceleration (0.025 vs. 0.036).

For physical evaluation (Table **??**), PINO significantly reduces overlap (0.031 vs. 0.672) and penetration volume (1375.24 vs. 15622.16 cm³) while maintaining natural motion dynamics. The boundary smoothness penalty proves effective at creating seamless transitions between motion segments.

5.5. Ablation Studies

Table **??** analyzes the contribution of individual penalty terms in our optimization framework. The results demonstrate the complementary nature of different constraints:

Root Position Control ($\mathcal{L}_{\text{root}}$): Eliminates positional errors ($1.0 \rightarrow 0.0$) but has limited impact on other metrics, confirming its role in precise spatial control.

Overlap Avoidance ($\mathcal{L}_{\text{overlap}}$): Completely eliminates overlap ($0.292 \rightarrow 0.0$) with minimal impact on other constraints, demonstrating effective physical constraint enforcement.

Region Constraints ($\mathcal{L}_{\text{region}}$): Dramatically reduces region violations ($0.958 \rightarrow 0.043$) while maintaining overlap elimination, enabling precise spatial control.

Orientation Control ($\mathcal{L}_{\text{orientation}}$): Significantly improves facing direction accuracy ($1.0 \rightarrow 0.208$) while maintaining other improvements, enabling directional interaction control.

Fig. **??** visualizes the effect of different penalty combinations, showing how each constraint contributes to more physically plausible and semantically meaningful interactions.

The ablation study confirms that all proposed penalty terms contribute meaningfully to the final result, with each addressing a specific aspect of physical or semantic plausibility. The modular nature of our approach allows practitioners to selectively apply constraints based on application requirements.

6. Limitations

While PINO demonstrates significant improvements in multi-person interaction generation, several limitations should be acknowledged:

Computational Overhead. The noise optimization process requires iterative gradient descent for each new character, which increases computational cost compared to direct sampling from pretrained models. The optimization typically requires 100 iterations per character, resulting in longer generation times for large groups.

Sequential Composition Artifacts. Although PINO's sequential approach enables scalability, it may occasionally produce slight inconsistencies in global group dynamics that concurrent generation methods might better capture. The choice of pivot character can influence the quality of subsequent additions.

Dependency on Base Model Quality. PINO's performance is fundamentally limited by the quality of the underlying two-person diffusion model. Poor base model performance in specific interaction types will propagate to the final multi-person result.

Hyperparameter Sensitivity. The penalty weights require careful tuning for optimal results. While we provide reasonable defaults, different interaction types or datasets may benefit from adjusted weight configurations.

Limited Long-Term Dependencies. For very long sequences, the motion inpainting approach may struggle to maintain consistent global narrative or character personalities across extended durations, though local temporal coherence is well preserved.

7. Conclusion

We have presented PINO (Person-Interaction Noise Optimization), a training-free framework for generating physically coherent multi-person interactions of arbitrary size and duration. By decomposing complex group dynamics into sequential pairwise interactions and employing physics-aware noise optimization, PINO addresses fundamental challenges in multi-person motion generation while maintaining the flexibility and quality of pretrained diffusion models.

Our key contributions include: (1) A scalable composition framework that leverages pretrained two-person models without requiring additional training, (2) Physics-aware optimization that eliminates overlap and enables fine-grained spatio-temporal control, (3) Per-pair prompt customization that allows nuanced role assignment within groups, and (4) Long-duration generation through motion inpainting with smooth transitions.

Extensive experiments demonstrate that PINO achieves significant improvements in physical plausibility—reducing overlap to near zero and decreasing penetration volume by over 90%—while maintaining motion quality comparable to state-of-the-art methods. The method scales effectively to multi-person scenarios and supports extended sequences, providing a practical solution for applications in animation, gaming, and robotics.

Future work could explore more sophisticated global coherence mechanisms for very large groups, automatic hy-

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perparameter adaptation for different interaction types, and integration with physics simulators for even more robust physical constraint enforcement. The modular nature of PINO's design makes it amenable to such extensions while preserving its core advantages of training-free operation and fine-grained control.

PINO represents a significant step toward practical and controllable multi-person motion generation, offering a framework that balances physical realism, semantic coherence, and computational efficiency. We believe this work will facilitate broader adoption of AI-generated human motions in real-world applications requiring complex multi-agent interactions.

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